THIRUVENKADAPRASATH.S

192312016

10	Write a Python Program to Implement Expectation & Maximization Algorithm

CSV FILE

Value

1.2

1.5

1.7

1.1

1.9

1.4

6.8

7.1

6.9

7.5

7.2

6.6

PROGRAM

```
import numpy as np
import pandas as pd


def gaussian(x, mean, var):
    return (1 / np.sqrt(2 * np.pi * var)) * np.exp(
        -((x - mean) ** 2) / (2 * var)
    )
```

```
df = pd.read_csv("em_algorithm_data.csv")
X = df["Value"].values
n_samples = len(X)

m1, m2 = 2.0, 5.0
v1, v2 = 1.0, 1.0
w1, w2 = 0.5, 0.5

iterations = 20

for i in range(iterations):
    pdf1 = gaussian(X, m1, v1)
    pdf2 = gaussian(X, m2, v2)

    gamma1 = (w1 * pdf1) / (w1 * pdf1 + w2 * pdf2)
    gamma2 = (w2 * pdf2) / (w1 * pdf1 + w2 * pdf2)

    n1 = np.sum(gamma1)
    n2 = np.sum(gamma2)

    m1 = np.sum(gamma1 * X) / n1
    m2 = np.sum(gamma2 * X) / n2

    v1 = np.sum(gamma1 * (X - m1) ** 2) / n1
    v2 = np.sum(gamma2 * (X - m2) ** 2) / n2

    w1 = n1 / n_samples
    w2 = n2 / n_samples

print("--- Estimated Parameters after Convergence ---")
```

```python
print(f"Cluster 1 -> Mean: {m1:.4f}, Variance: {v1:.4f}, Weight: {w1:.4f}")
print(f"Cluster 2 -> Mean: {m2:.4f}, Variance: {v2:.4f}, Weight: {w2:.4f}")

cluster_assignments = np.where(gamma1 > gamma2, 1, 2)
df["Assigned_Cluster"] = cluster_assignments

print("\n--- Data Point Cluster Assignments ---")
print(df)
```

OUTPUT